

MD. MUTASIM BILLAH ABU NOMAN AKANDA

PhD Applicant (Fall 2027) • Document AI and Low-Resource OCR

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Research Interests

Document AI and OCR for low-resource languages (layout analysis, handwriting recognition, layout-aware evaluation); efficient multimodal / VLM adaptation under deployment budgets; biomedical image analysis (MRI/CT DICOM pipelines); reproducible multi-GPU training and metric design. Current focus: Bengali handwritten OCR with parse-success-gated text-word-layout scoring, sweep-guided model selection, and 4-GPU DDP workflows.

Education

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
B.Sc. Computer Science and Engineering Summer 2019 – Fall 2022
CGPA: 3.96/4.00 **Distinction:** Summa Cum Laude (top 1%; conferred May 2024)
Honors: Vice Chancellor's Honors List; Dean's List Scholarship (3×)
Relevant coursework: Artificial Intelligence, Digital Image Processing, Algorithms, Data Structures, Statistics and Probability, Software Engineering

Publications

- **Akanda, M.B.A.N.*, Ahmed, M.*, Rabby, A.S.A., & Rahman, F.:** *Optimum Deep Learning Method for Document Layout Analysis in Low Resource Languages*. ACM Southeast Conference (ACM SE'24), 199–204. doi:10.1145/3603287.3651184. *Equal contribution.
- **Akanda, M.B.A.N., Prodhan, M., Sarwar, S., Raatul, A.M., Paul, B.:** *Voice Controlled Home Automation with Cloud-Based Environment Monitoring System*. ICTCS 2022, Lecture Notes in Networks and Systems, vol. 623, Springer, Singapore. doi:10.1007/978-981-19-9638-2_21

Research and Professional Experience

- **Apurba Technologies Ltd.** Dhaka, Bangladesh
Senior Machine Learning Engineer Sep 2025 – present
 - **Current research project (ongoing):** Leading Bengali handwriting OCR fine-tuning with a reproducible workflow: strict JSONL validation, W&B sweep screening, and 4-GPU DDP full training for final model selection.
 - **Objective-aligned evaluation:** Using a joint text-word-layout optimization target ('eval_text_word_layout_score') with parse-success gating to prioritize both textual correctness and structural fidelity.
 - **Training-system rigor:** Implemented sweep-equivalent global batch behavior across GPU counts and DDP-safe OCR metrics controls to preserve comparability between screening runs and full training.
 - **Research relevance:** Converting production constraints (layout variance, OCR noise, memory limits) into measurable experiments on robustness, efficiency, and deployment reliability.
- **Apurba Technologies Ltd.** Dhaka, Bangladesh
Machine Learning Engineer Mar 2023 – May 2024
 - **Low-resource document layout analysis:** Built and evaluated YOLOv8-based layout models for Bengali document structures (85%+ internal accuracy), supporting OCR quality improvements in deployment.
 - **Inference optimization:** Implemented 8-bit quantization and measured 2× faster inference with reduced memory usage, enabling broader operational deployment.
 - **Research dissemination:** Co-first author on ACM SE'24 Bengali DLA study (BADLAD); co-author on Springer ICTCS 2022 paper.
- **Pixalate Inc.** Remote (California, USA)
Data Scientist Mar 2025 – Aug 2025
 - **Web-scale ML systems:** Designed AI-assisted ad-network detection workflows over large crawling pipelines (2M+ sites), combining learned and deterministic methods.
 - **Methodological trade-off analysis:** Used hybrid detection strategies to balance precision, throughput, and infrastructure cost in production settings.
- **Green Pants Studio** Remote (Texas, USA)
AI Engineer Jun 2024 – May 2025

- **Applied ML for valuation and mapping:** Developed ML-supported product valuation and data mapping workflows for large eCommerce streams.
- **Pipeline quality and reliability:** Improved scraper throughput and deployment reliability through modular services and containerized delivery.

Teaching Experience

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
Teaching Assistant, Department of Computer Science and Engineering Feb 2021 – Jan 2023
 - **Courses supported:** Introduction to Programming, Object-Oriented Programming, Artificial Intelligence, Software Engineering
 - **Instructional duties:** Led laboratory sessions and tutorials for cohorts of **30+** students; prepared lab materials and evaluated assignments/projects.
 - **Mentoring:** Guided students on programming and introductory AI/ML projects during office hours and practical sessions.

Leadership

- **ULAB Computer Programming Club** Dhaka, Bangladesh
President 2022 – 2023
 - **Contest organization:** Led 4 UCPC competitive programming contests (2022–2023), including Take Off Summer 2022 (**40+** participants).
 - **Problem setting:** Lead problem setter, Take Off Summer 2022 (**6**-problem set; ULAB official announcement).

Biomedical Imaging (Kaggle / RSNA)

- **RSNA-MICCAI Brain Tumor Radiogenomic Classification:** Built PyTorch EfficientNet3D pipelines on mpMRI DICOM (FLAIR, T1w, T1wCE, T2w) for MGMT promoter methylation classification; middle-slice volume stacking and DICOM preprocessing with VOI LUT. Kaggle
- **RSNA 2023 Abdominal Trauma Detection:** Explored CT trauma detection with PyTorch 2.5D EfficientNet CNN (5-fold CV), 3D R3D-18 on DICOM slices, KerasCV CNN training, and DICOM-to-3D tensor preprocessing. Kaggle

Selected Projects (Research-Relevant)

- **SenseScan (Bengali handwritten OCR):** Pipeline with EAST/LANMS detection, CRNN recognition, and reading-order assembly; designed for experimentation in low-resource OCR settings. GitHub
- **Voice Healthcare Agent:** Multimodal conversational pipeline (FastAPI + Next.js + local voice stack) useful for studying latency/reliability trade-offs in speech-enabled LLM systems. GitHub
- **GradConnectAI:** Evidence-based matching system for graduate supervision discovery; relevant to ranking/retrieval experimentation in academic recommendation settings. GitHub

Reproducibility and Tooling

- **Experiment support:** Use MLflow/Weights & Biases style tracking and test-driven workflows (pytest/Vitest) to improve repeatability and regression control in ML pipelines.
- **Deployment-aware reproducibility:** Maintain API-first project structures (FastAPI, Docker, CI) to align research prototypes with executable, verifiable systems.
- **Data and evaluation mindset:** Prioritize clear evaluation criteria, metric definitions, and error-analysis loops when moving from prototype to production conditions.

Technical Skills

- **Languages:** Python, TypeScript, SQL
- **ML/AI:** PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, quantization workflows
- **Medical imaging:** pydicom, DICOM preprocessing, EfficientNet3D, 3D/video CNNs (R3D-18), KerasCV
- **LLM/NLP:** RAG pipelines, vector search (pgvector/ChromaDB), vLLM, Ollama
- **Systems:** FastAPI, Docker, PostgreSQL, Redis, SQLite, async crawling and data pipelines
- **Experimentation:** MLflow, Weights & Biases, pytest, Vitest

Honors and Awards

- **Summa Cum Laude:** Top 1% of graduating class (May 2024)
- **Vice Chancellor's Honors List:** Scholarship recipient
- **Dean's List Scholarship:** Awarded 3 times
- **Programming contests:** Problem setter and award recipient in ULAB Take Off contests
- **ULAB Computer Programming Club:** President (2022–2023); organized 4 contests; Take Off Summer 2022 (40+ participants)